

Bend Theory of River Meanders

Part 1 — Linear Development

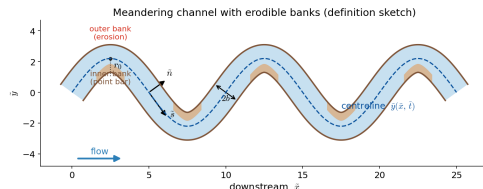
Ikeda, Parker & Sawai (1981), *J. Fluid Mech.* **112**, 363–377

A visual walkthrough

Why do rivers meander with a preferred wavelength, and why do the bends crawl downstream?

Rivers meander — regularly, and on the move

- Alluvial rivers rarely run straight: they wind in **quasi-regular bends**.
- Two robust field observations demand explanation:
 1. a **characteristic wavelength** λ set by the channel scale;
 2. bends **migrate downstream** and grow over years–centuries.
- Width stays roughly *constant*: outer-bank erosion is matched by inner-bank (point-bar) deposition.



The paper derives *both* the wavelength and the migration from first principles.

Two competing ideas: *bar* vs *bend*

Earlier “bar” theories

- Straight channel, **rigid banks**.
- Alternate bars form on the bed; flow winds *around* them.
- Bank deformation is *not allowed* — so a true meander cannot develop.

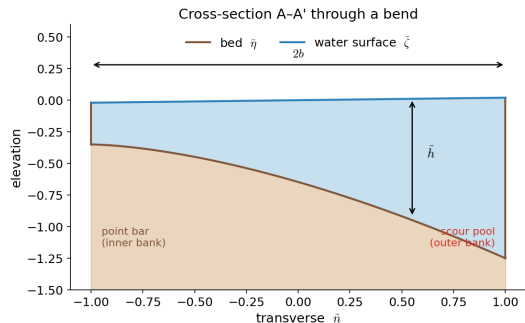
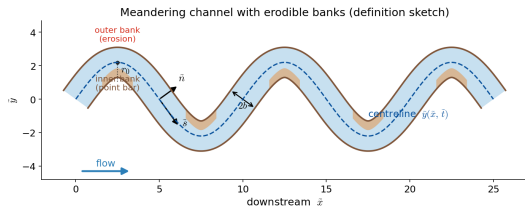
This paper’s “bend” theory

- **Erodible banks**: the bend itself is unstable.
- Curvature $\rightarrow$ secondary flow $\rightarrow$ selective bank erosion $\rightarrow$ growth *and* migration.
- Applies to alluvial, incised (bedrock/cave), and glacial meanders alike.

Goal: a **linear stability analysis** of a sinuous, erodible-bank channel — find the growth rate $\alpha_0(k)$ and migration speed $c_0(k)$ of a small meander perturbation.

1. **Setup:** geometry, assumptions, the depth-averaged equations.
2. **Closures:** secondary flow (the parameter A) and a bank-erosion law.
3. **The bend equation:** one linear PDE for the centreline $y(x, t)$.
4. **Dispersion relation:** growth rate, migration speed, the selected wavelength.
5. **The mechanism:** the curvature–velocity *phase lag*.
6. **Results:** wavelength selection, downstream migration, comparison with data.

Geometry and assumptions



Intrinsic coordinates: $\tilde{s}$ downstream, $\tilde{n}$ transverse; centreline $\tilde{y}(\tilde{x}, \tilde{t})$ with curvature $\tilde{\mathcal{C}}$.

Three assumptions

1. Constant width $2b$ (erosion $\approx$ deposition).
2. Gentle bends: $\nu = b/r_0 \ll 1$ (expand to $O(\nu^2)$).
3. Quasi-steady flow.

Small-amplitude (*linear*) limit: $\delta_0 = k\varepsilon \ll 1$, with $\varepsilon = y_0/H_0$, $k = 2\pi H_0/\lambda$.

Momentum $(\tilde{s}, \tilde{n})$ and continuity for a sinuous channel:

$$\tilde{u} \partial_{\tilde{s}} \tilde{u} + \tilde{v} \partial_{\tilde{n}} \tilde{u} + \tilde{C} \tilde{u} \tilde{v} = -g \partial_{\tilde{s}} \tilde{\zeta} - \tilde{\tau}_s / (\rho \tilde{h}), \quad (1a)$$

$$\tilde{u} \partial_{\tilde{s}} \tilde{v} + \tilde{v} \partial_{\tilde{n}} \tilde{v} - \tilde{C} \tilde{u}^2 = -g \partial_{\tilde{n}} \tilde{\zeta} - \tilde{\tau}_n / (\rho \tilde{h}), \quad (1b)$$

$$\tilde{C} \tilde{v} \tilde{h} + \partial_{\tilde{n}} (\tilde{v} \tilde{h}) + \partial_{\tilde{s}} (\tilde{u} \tilde{h}) = 0. \quad (1c)$$

Bed stress via a (constant) friction coefficient C_f :

$$\tilde{\tau}_s = \rho C_f \tilde{\mathcal{W}} \tilde{u}, \quad \tilde{\tau}_n = \rho C_f \tilde{\mathcal{W}} \tilde{v}, \quad \tilde{\mathcal{W}} = (\tilde{u}^2 + \tilde{v}^2)^{1/2}.$$

Zeroth order = uniform flow

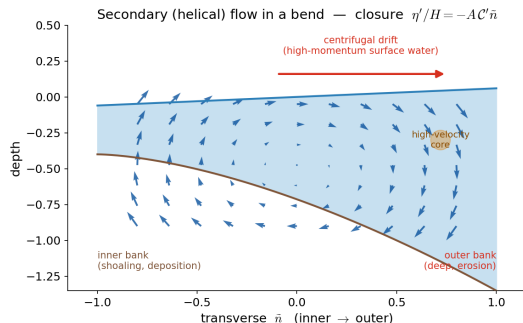
$$C_f U^2 = g H I, \quad U H = q_w \quad (\text{friction balances gravity}).$$

Closure 1 — secondary flow and the parameter A

The 2-D St Venant system cannot itself capture the **helical circulation** in a bend. A 3-D closure (Engelund 1974; ...) gives, to $O(\nu)$:

$$\boxed{\frac{\eta'}{H} = -AC' \tilde{n}} \quad (6)$$

- Transverse bed slope (and shifted velocity core) $\propto$ curvature.
- $A = O(1)$; field data (45 bends): $A \simeq 2.89$ for alluvial rivers.
- $A = 0$ for a laterally flat / incised bed.



Near-bank velocity — where the lag is born

Substituting the closures into the transverse balance yields a **forced, first-order** equation for the near-bank velocity $u_b = (u')_{\tilde{n}=b}$ (linear limit $\chi = \gamma = 1$ of Eq. 10):

$$\boxed{\frac{\partial u_b}{\partial s} + 2C_f u_b = b^* \left[-\frac{\partial \mathcal{C}}{\partial s} + C_f (A + F^2) \mathcal{C} \right]} \quad F = \frac{U_0}{\sqrt{gH_0}}.$$

- The forcing mixes curvature $\mathcal{C}$ and its along-stream gradient $\partial_s \mathcal{C}$.
- A first-order operator with friction damping $2C_f \Rightarrow$ the response u_b is **phase-shifted** relative to $\mathcal{C}$: it peaks *downstream* of the bend apex.
- That shift is the engine of everything that follows.

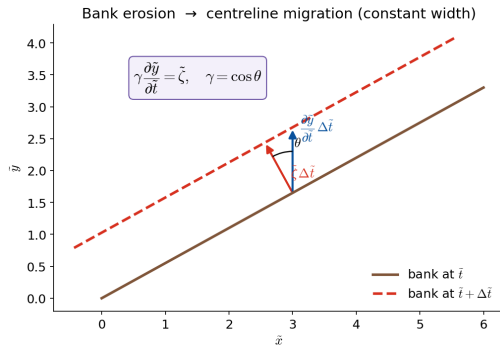
Closure 2 — the bank-erosion law

Bank migrates at a rate set by the *excess* near-bank velocity:

$$\gamma \partial_{\tilde{t}} \tilde{y} = \tilde{\zeta}, \quad \gamma = \cos \theta \quad (11)$$

$$\tilde{\zeta} = \tilde{\zeta}(U) + \underbrace{E(U)}_{>0} u_b(\tilde{s}, b) \quad (12)$$

- Faster-than-average near-bank flow $\Rightarrow$ erosion; slower $\Rightarrow$ deposition opposite.
- Constant width $\Rightarrow$ the whole *centreline* shifts.
- The erodibility slope $e = \frac{1}{E} \frac{dE}{d\chi}$ matters only at nonlinear order (Part 2).



Assembling the linearized bend equation

Combining the momentum balance, secondary-flow closure and bank-erosion law, and transforming to Cartesian coordinates ($\partial_s = \gamma \partial_x$, $\mathcal{C} = -\gamma^3 y_{xx}$), the small-amplitude limit collapses to a single PDE for the centreline $y(x, t)$:

$$\boxed{\frac{\partial^2 y}{\partial x \partial t} + 2C_f \frac{\partial y}{\partial t} = \frac{\partial^3 y}{\partial x^3} - C_f (A + F^2) \frac{\partial^2 y}{\partial x^2}} \quad (16)$$

- y_{xt}, y_t : rate of change of curvature and lateral migration.
- y_{xxx} : dispersive term from the curvature-gradient forcing.
- $-C_f(A + F^2)y_{xx}$: the secondary-flow + Froude driving of growth.

Normal modes $\Rightarrow$ the dispersion relation

Seek a travelling, growing bend (their Eq. 17):

$$y = \varepsilon e^{\alpha_0 t} \cos(kx - \omega_0 t), \quad c_0 = \omega_0/k \text{ (migration speed).}$$

Substituting into (16) and splitting real/imaginary parts gives

$$\omega_0 = \frac{C_f k^3 (2 + A + F^2)}{k^2 + 4C_f^2}, \quad (18a)$$

$$\alpha_0 = \frac{2C_f^2(A + F^2)k^2 - k^4}{k^2 + 4C_f^2}. \quad (18b)$$

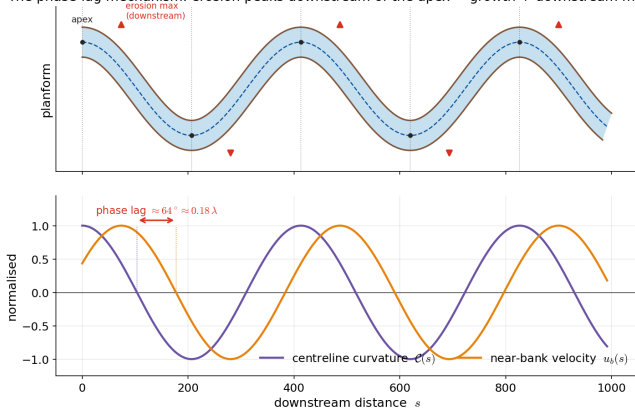
Two immediate readings

Instability ($\alpha_0 > 0$): $k < k_c = \sqrt{2} C_f \sqrt{A + F^2}$ (Eq. 19)

Fastest growth: $k_{OM} = \beta C_f$, $\beta^2 = 4\sqrt{1 + \frac{1}{2}(A + F^2)} - 4$ (Eqs. 20–21)

The heart of it: the curvature–velocity phase lag

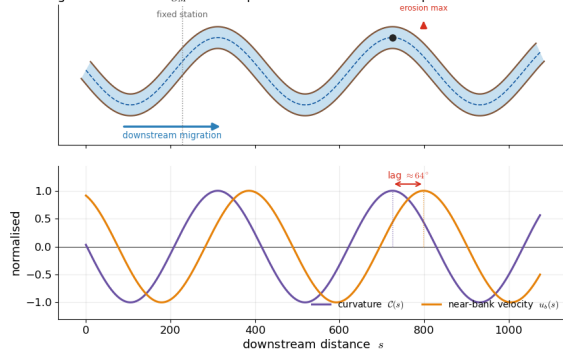
The phase-lag mechanism: erosion peaks downstream of the apex $\rightarrow$ growth + downstream migration



Curvature peaks at the apex; near-bank velocity (and erosion) peaks $\approx 64^\circ \approx 0.18\lambda$ **downstream** $\Rightarrow$ the outer bank just past each apex erodes fastest $\Rightarrow$ **growth + downstream migration**.

The mechanism, in motion

Phase-lag mechanism at k_{OM} : erosion peaks downstream of the apex $\Rightarrow$ the bend train migrates

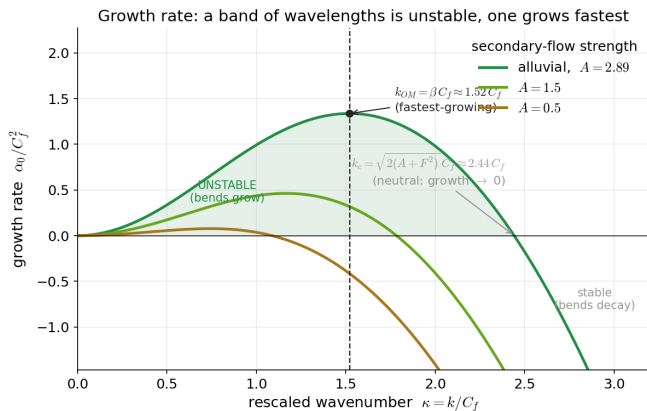


A travelling bend at k_{OM} :

- curvature $C(s)$ and near-bank velocity $u_b(s)$ travel together,
- separated by the fixed $\sim 64^\circ$ lag,
- so the erosion maximum always rides the downstream flank.

► animation:
figures/phase_lag.mp4

Growth rate: a band is unstable, one wavelength wins

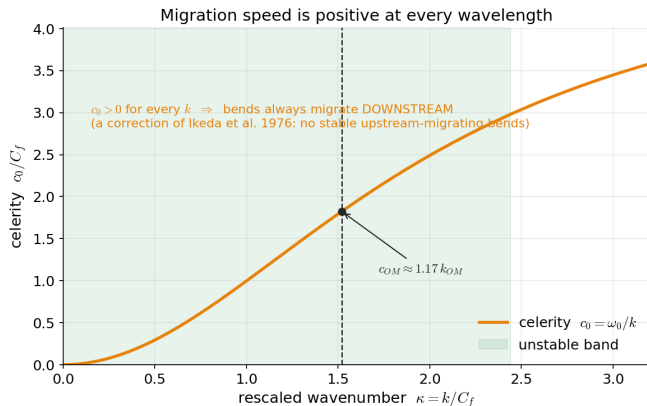


In rescaled $\kappa = k/C_f$ the curves are **C_f -independent**:

$$\frac{\alpha_0}{C_f^2} = \frac{2(A + F^2)\kappa^2 - \kappa^4}{\kappa^2 + 4}.$$

- Long waves ($\kappa \rightarrow 0$): too little forcing.
- Short waves ($\kappa > \kappa_c$): friction wins.
- Peak at $\kappa_{OM} = \beta$.

Bends always migrate downstream



Since $\omega_0 > 0$ for every k , the celerity

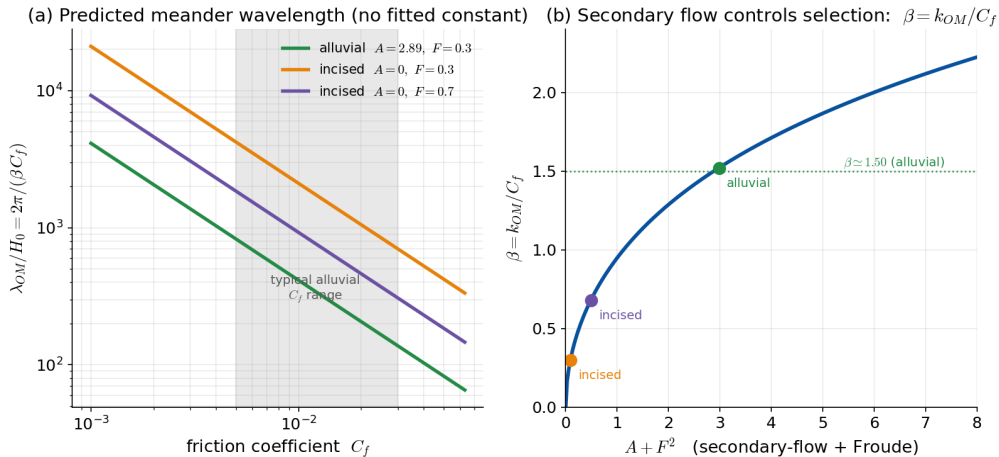
$$c_0 = \frac{C_f k^2 (2 + A + F^2)}{k^2 + 4C_f^2} > 0$$

is **always positive**.

- Bends march *downstream* regardless of Froude number.
- Corrects the earlier heuristic (Ikeda et al. 1976) that stable bends migrate upstream.

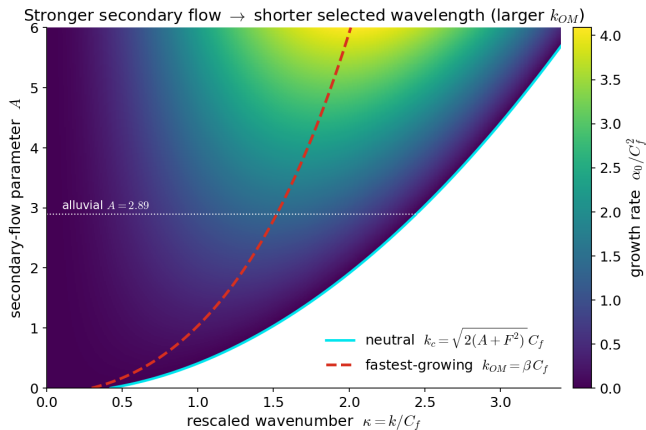
Wavelength selection — with no fitted constant

Wavelength selection: friction sets the scale, secondary flow sharpens it (theory curves, Eqs. 20–21, 24a)



Alluvial ($A \simeq 2.89, F^2 \ll A$): $k_{OM} = 1.50 C_f$, $\alpha_{OM} = 0.564 k_{OM}^2$, $\omega_{OM} = 1.17 k_{OM}^2$ (Eqs. 24). Incised ($A = 0$): $k_{OM} \simeq C_f F$ (Eq. 28) $\Rightarrow$ longer wavelengths.

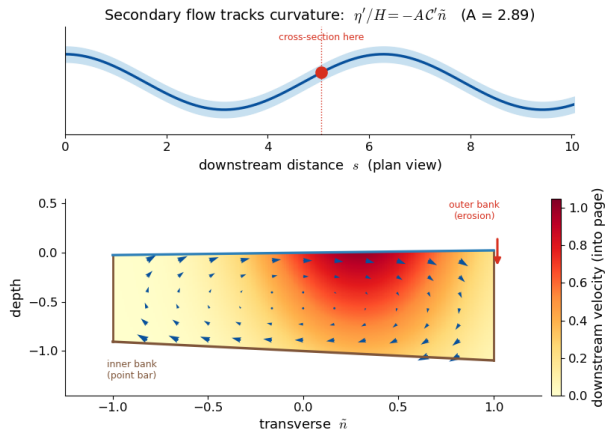
How secondary flow reshapes the selection



Growth rate over (κ, A) at fixed F :

- stronger secondary flow (larger A)
 $\Rightarrow$ **larger** k_{OM} $\Rightarrow$ *shorter* selected wavelength;
- the unstable band widens with A ;
- alluvial rivers sit at $A \approx 2.89$.

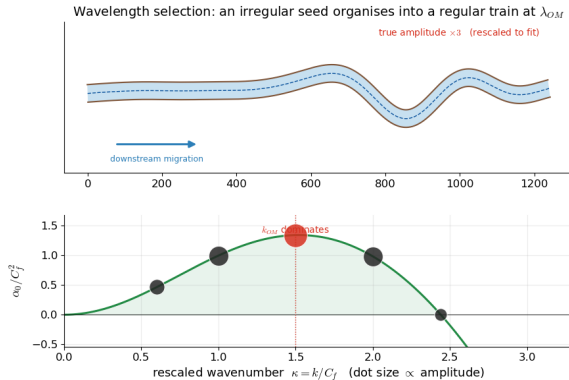
Secondary flow, animated



Sweeping along the meander, the helical cell, super-elevation, tilted bed and outer-bank velocity core all track the local curvature — and **reverse** through each inflection.

► animation:
figures/secondary_flow.mp4

From a jumble to a regular train



Seed the channel with many wavenumbers; each evolves by its own $e^{\alpha_0(k)t} \cos(kx - \omega_0 t)$.

- Sub-cutoff modes grow, the rest decay.
- The k_{OM} mode overtakes $\Rightarrow$ a **regular meander train**.

► animation:
[figures/meander_evolution.mp4](#)

Comparison with data (theory curves)

- The paper tests $k_{OM} = 1.50 C_f$ against **158 laboratory + 73 natural** alluvial reaches (their Figs. 4–5): rough agreement over *five decades* of wavelength.
- Crucially, bend theory carries **no adjustable constant**, unlike the competing bar-theory relation $k_{OM} = a C_f^{1/4} l^{1/4}$ (with fitted $a \simeq 0.707$).
- Incised cases ($A = 0$): rill meanders follow bend theory (Eq. 26a); supraglacial meanders look more like bar instability. Cave-stream planforms (Gardner's Gut) show the predicted *downstream* migration.

Here we plot the exact prediction curves rather than reproduce the scatter — the theory has a definite scaling to compare against.

- Linear theory holds only while $\delta_0 = k\varepsilon \ll 1$ — the *early* growth of small bends.
- As bends fatten, geometric (δ_0^3) and dynamic nonlinearities re-enter: amplitude saturation, skewing, and eventually **neck cutoff**.
- **Part 2** (Parker, Sawai & Ikeda 1982) carries the expansion to finite amplitude to capture these.

1. A sinuous, erodible-bank channel is **linearly unstable**: straight channels spontaneously meander.
2. The mechanism is a **phase lag** — near-bank velocity/erosion peaks $\sim 0.18\lambda$ downstream of each apex.
3. This yields a clean **dispersion relation**: a preferred wavelength $k_{OM} = \beta C_f$ and downstream migration $c_0 > 0$ for all k .
4. For alluvial rivers $\beta \simeq 1.50$; the prediction matches data over five decades *with no fitted constant*.
5. Secondary flow (parameter A) both drives growth and sets the wavelength.

One idea to remember: a small lag between where a river bends and where it flows fastest is enough to make meanders *grow* and *march downstream*.

- Ikeda, Parker & Sawai (1981) *Bend theory of river meanders. Part 1*. J. Fluid Mech. **112**, 363–377. (this talk)
- Parker, Sawai & Ikeda (1982) ...*Part 2. Nonlinear deformation*. J. Fluid Mech.
- Engelund (1974) *Flow and bed topography in channel bends*. J. Hydraul. Div. ASCE **100**.
- Ikeda, Hino & Kikkawa (1976); Kikkawa, Ikeda & Kitagawa (1976) — secondary-flow closures.
- Zimmerman & Kennedy (1978) *Transverse bed slopes in curved alluvial streams*.
- Hooke (1975); Leopold, Wolman & Miller (1964) — field/lab data.

Figures & animations generated from the verified linear relations in `ikeda_lib.py`.